

No Borrowed Recovery

The 2025 Teacher-Workforce Data for the United States

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In brief. England's 2025 school-workforce data showed teacher retention recovering — a recovery the Education Policy Institute called *borrowed*, because it was lent to schools by a weak graduate labour market rather than earned by policy. This note runs the same reading of the 2025 data for the United States, using the Current Population Survey (CPS) in place of England's School Workforce Census. The American answer is starker: the U.S. has no recovery to give back, because it never borrowed one.

- In 2025 the stock of full-time-qualified school teachers fell by about 3.7 per cent, the first clear post-pandemic decline; in the 2024→25 panel, exits (20.3%) clearly outpaced entries (16.6%).
- Teacher attrition is off its 2022 peak (18.2%) but stuck at 17.6% in 2024→25 — roughly three points above its pre-COVID norm (14.5%). Persistent leavers: 15.4%.
- The outside labour market has cooled exactly as in England: private quits are back to 2015 levels. Historically teacher attrition tracks quits ($r = +0.60$); today it runs *two points above* what that relationship predicts.
- Real teacher pay is 5.6 per cent below its 2010 level, and has fallen from 87 to 79 cents per graduate-median dollar. England cut teacher pay too — but its market slump still delivered retention. The U.S. got the pay cut without the retention.
- The inflow now runs on career-changers: the median new entrant is 42 years old, three quarters arrive from another job, and fewer than one in five is under 30 — the American twin of England's returner-dependent primary schools.

1. England's borrowed recovery, and the American question

In June 2026 the Education Policy Institute published its reading of England's 2025 School Workforce Census under a memorable title: *A borrowed recovery* (Education Policy Institute, 2026). The English numbers were, on their face, good news — attrition falling, early-career retention near a decade high,

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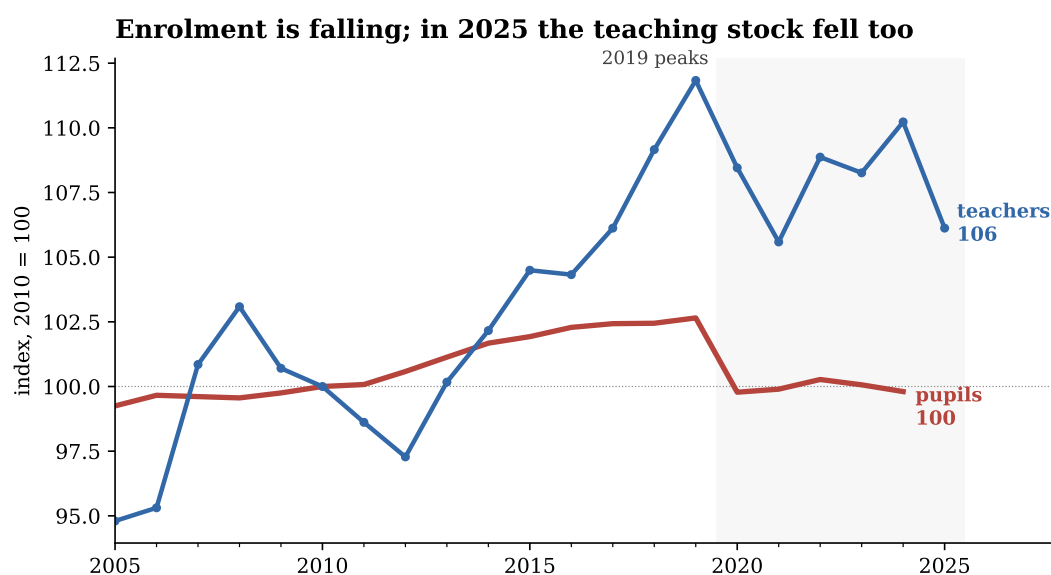


Figure 1: Public-school pupils and full-time-qualified teachers, indexed to 2010.

Teachers: CPS monthly cross-sections, employed K–12 teachers (occupations 2310–2330) holding at least a bachelor’s degree, annual average of weighted monthly stocks (2025: eleven months — no CPS was collected in October 2025 during the federal shutdown; on a like-for-like eleven-month basis the 2024 stock is 4.52M, so the 2025 decline is not an artefact). Pupils: NCES Digest of Education Statistics table 203.10, fall enrolment, public schools. CPS stocks are survey estimates and carry sampling error of roughly ± 1 –2 per cent.

recruitment up. EPI’s point was that almost none of this was earned: pupil rolls are falling, the graduate labour market is weak, and teacher pay is still nine per cent below its 2010 value in real terms. The recovery was lent to schools by the economy, and what the economy lends it can recall.

This note asks the same question of the United States, replicating each element of the EPI analysis that American data permit. The U.S. has no census of teachers; what it has instead is the Current Population Survey, whose rotation design lets each teacher be re-observed exactly twelve months later. From the 2005–2025 monthly files I link 148,390 full-time K–12 teachers with a bachelor’s degree or higher and follow each for a year, giving attrition, entry, destination and pay series that run through the 2024→2025 school year — the same vintage as EPI’s data. (The companion paper, *Who Leaves Teaching, and Why?*, documents the linkage; the data note at the end lists what the CPS cannot replicate and what stands in for it.)

The answer does not mirror England’s. It inverts it.

2. Fewer pupils — and now fewer teachers

Start with the demand side, where the two countries rhyme. England’s pupil rolls have been falling since 2019; American public-school enrolment peaked in autumn 2019 at 50.8 million and has since lost 1.4 million pupils (2.8 per cent), with NCES projecting further decline as smaller cohorts arrive. Through 2024, the American teaching stock ignored this: it recovered from the 2021 trough and hovered near its 2019 peak, so pupil–teacher ratios quietly improved — the same demography-driven easing EPI documents for England (Figure 1).

In 2025 that changed. The teaching stock fell from 4.54 to 4.37 million — 3.7 per cent, the sharpest one-

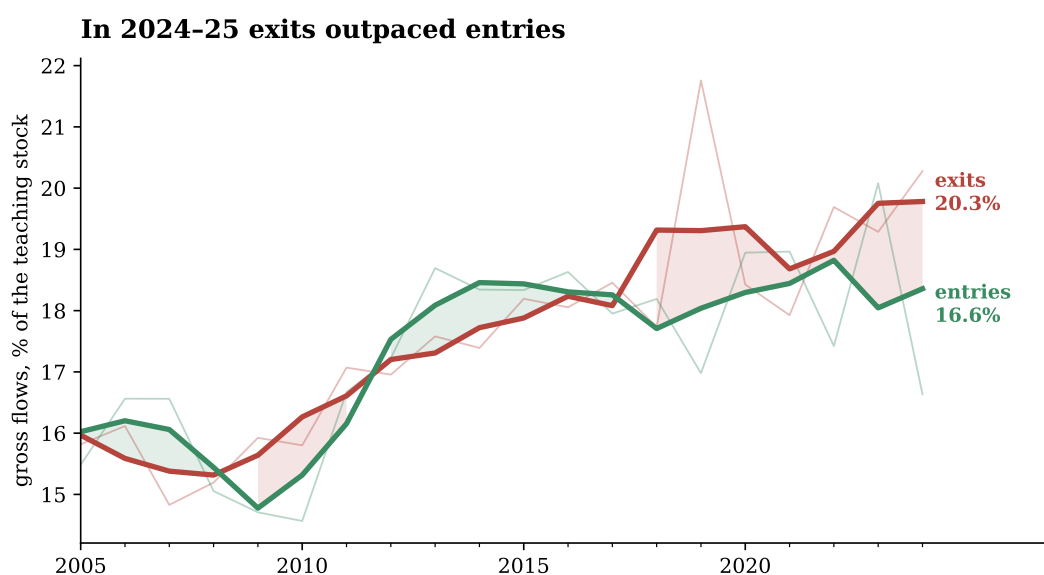


Figure 2: Gross entry and exit rates of qualified teachers, linked CPS year-pairs.

Entries: persons teaching (BA+) at $t+12$ who were not teaching at t ; exits: teachers (BA+) at t not teaching at $t+12$; each expressed on the corresponding teaching stock. Thick lines: three-year centred moving averages; pale lines: annual values; end labels give the raw 2024→25 rates.

year drop in the series outside the pandemic. England’s workforce also shrank in 2025 (by 1,910 FTE), but there the contraction was demand-led: fewer pupils, smaller training targets. The American contraction is supply-led. In the 2024→25 linked panel, 20.3 per cent of teachers left within twelve months while entries replaced only 16.6 per cent of the stock (Figure 2): the workforce is not being trimmed; it is leaking.

Who fills the gap that entries do cover? Not new graduates. The median teacher entering between 2024 and 2025 was *42 years old*; 75 per cent arrived directly from another job, 18 per cent from outside the labour force, and fewer than one in five was under 30. EPI’s most-cited finding this year is that returners now outnumber newly qualified teachers in England’s primary schools. The American inflow has the same face: mid-career adults circling back into classrooms, not a pipeline of the young.

3. Off the peak, but not back to normal

England’s wastage rate has fallen to near its pandemic low. The American series has not (Figure 3). Attrition averaged 14.5 per cent across 2005–18, spiked with COVID (the 2019 bar links teachers from 2019 into 2020) and again in the Great Resignation year of 2022 (18.2 per cent), then eased — but only to 16.9 in 2023→24, and it ticked *up* to 17.6 in 2024→25. The stricter measure that ignores brief exits — persistent leavers, out of teaching at $t+12$ and at every later interview — reads 15.4 per cent, also above its 2023 value. Churn is real but modest: about one in six of the 2024 leavers is back teaching within three months. However measured, the United States enters 2026 with attrition roughly three points above its old normal, at a moment when England celebrates retention near decade bests.

The age profile sharpens the point (Figure 4). Attrition among teachers in their thirties and forties has drifted up gently; among the under-30s it has climbed from about 14 per cent in the early 2010s to 17.2 per cent across 2022–24 and shows no easing. England’s recovery is strongest precisely where America’s problem is worst: EPI reports early-career retention near its best in a decade, while the youngest American

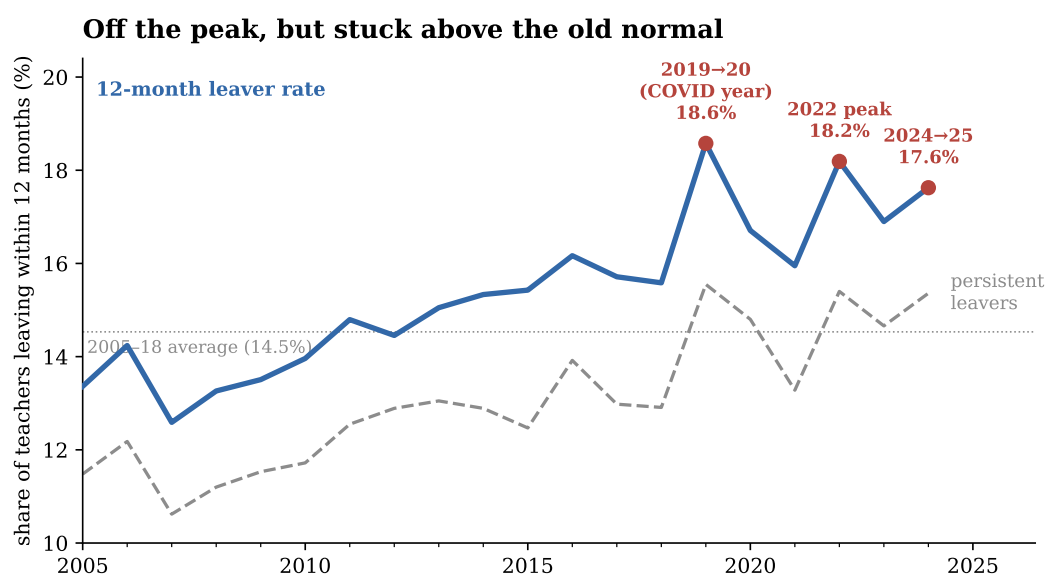


Figure 3: Teacher attrition, 2005–2024 base years.

Full-time K–12 teachers with BA+, followed twelve months in the linked CPS panel ($n = 148,390$; 4,913 in the 2024 base year, which lacks October links because no October 2025 CPS exists). Solid: share no longer teaching at $t+12$. Dashed: persistent leavers — no longer teaching at $t+12$ nor in any later observable interview.

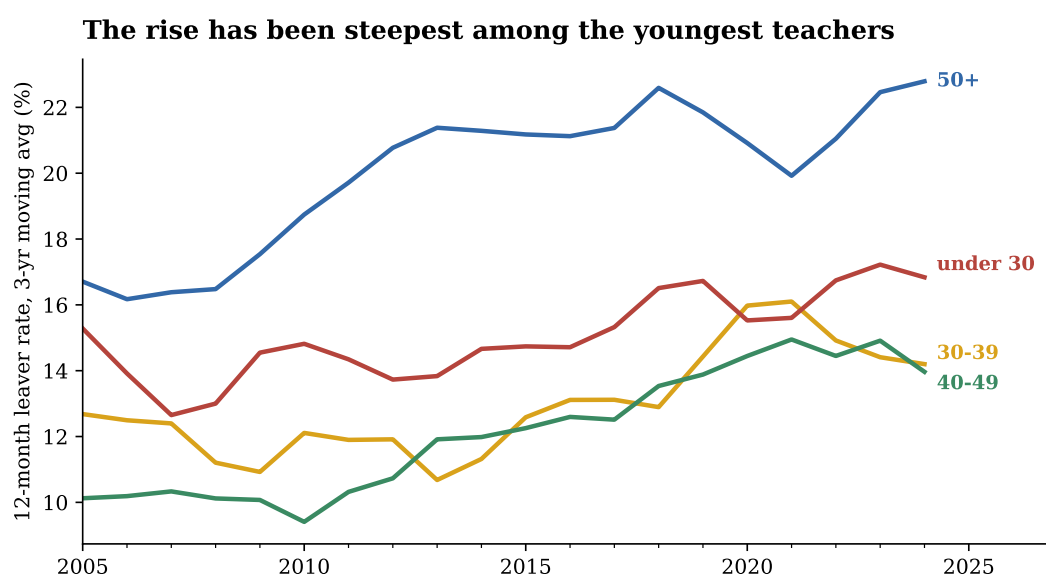


Figure 4: Attrition by age of teacher.

Same universe as Figure 3, split by age at the base interview; three-year centred moving averages. The 50+ line embeds retirements and is structurally high; the under-30 line is the one policy can read as early-career wastage.

teachers now leave at rates once reserved for the retirement-age group.

4. Pay: the lever that never moved

EPI's central warning is about pay: English teacher pay is 9.3 per cent below 2010 in real terms (the STRB says 13.9), so the moment the outside market recovers, the discount will bite. The American discount is

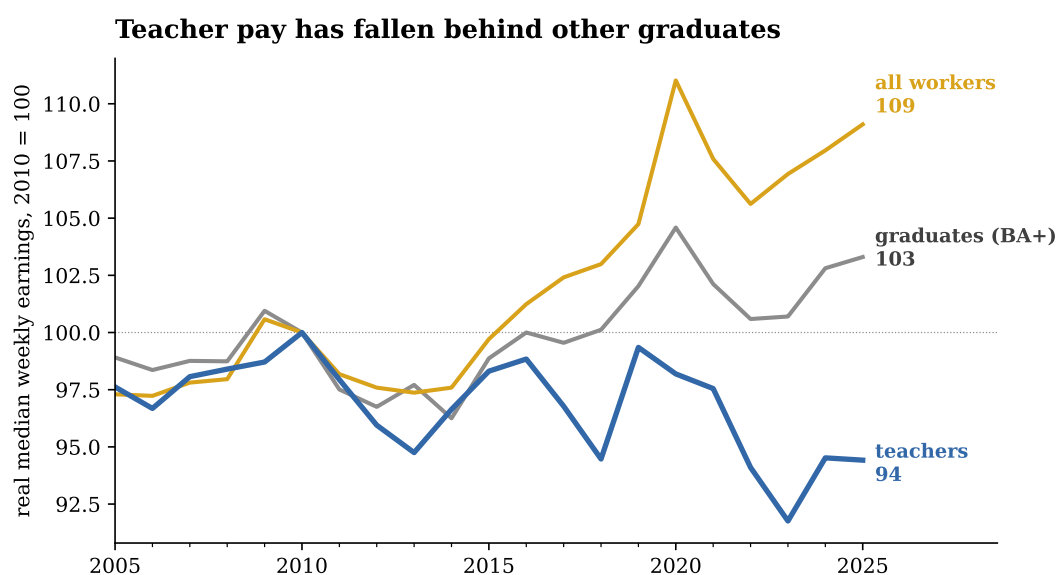


Figure 5: Real median weekly earnings: teachers, all graduates, all workers.

Teachers: weighted median usual weekly earnings of full-time K–12 teachers (BA+) in the panel’s outgoing-rotation interviews, deflated by CPI-U. Graduates: BLS/CPS median usual weekly earnings, bachelor’s degree and higher, 25+ (FRED LEU0252918500), same deflator. All workers: BLS real median usual weekly earnings, 16+ (FRED LEU0252881600). All series indexed to 2010 = 100.

of the same family (Figure 5). Real median weekly earnings of full-time teachers are 5.6 per cent below their 2010 level — \$1,380 against \$1,462 in 2025 dollars — while other degree-holders gained 3 per cent and the economy-wide median rose 9. Relative to the graduate median, teacher pay has slid from 87 to 79 cents on the dollar since 2010. This is the CPS-measured counterpart of the “teacher pay penalty” the (American) Economic Policy Institute tracks annually, and it widened exactly when retention needed it most: the inflation years 2021–23 cut teachers’ real weekly pay by nearly 6 per cent in two years.

5. The borrowed-recovery test

Here is the crux. EPI’s mechanism is that teacher retention is countercyclical: when outside options sour, teachers stay. The American data love this mechanism — across 2005–24, teacher attrition moves with the private-sector quits rate with a correlation of +0.60 (Figure 6): teachers quit when everyone quits, stay when everyone stays. And the American labour market has cooled on cue: private quits have fallen from 3.1 per cent at the 2022 frenzy to 2.3 per cent, JOLTS layoffs aside a market as calm as 2015’s.

So the United States ordered the same loan England took out — and the loan never arrived. At a 2.3 per cent quits rate, the historical relationship predicts teacher attrition of 15.6 per cent; the observed rate is 17.6. The 2024 point sits two full points above the line, the largest positive gap in the series after the COVID year (Figure 6, right panel). England’s weak market bought it retention it had not paid for; America’s equally weak market is buying nothing. Whatever pushed teacher attrition up after 2020 — the eroded pay position of Section 4, working conditions, the politicisation of the classroom — it is no longer the business cycle, because the cycle has already turned and attrition has not followed it down.

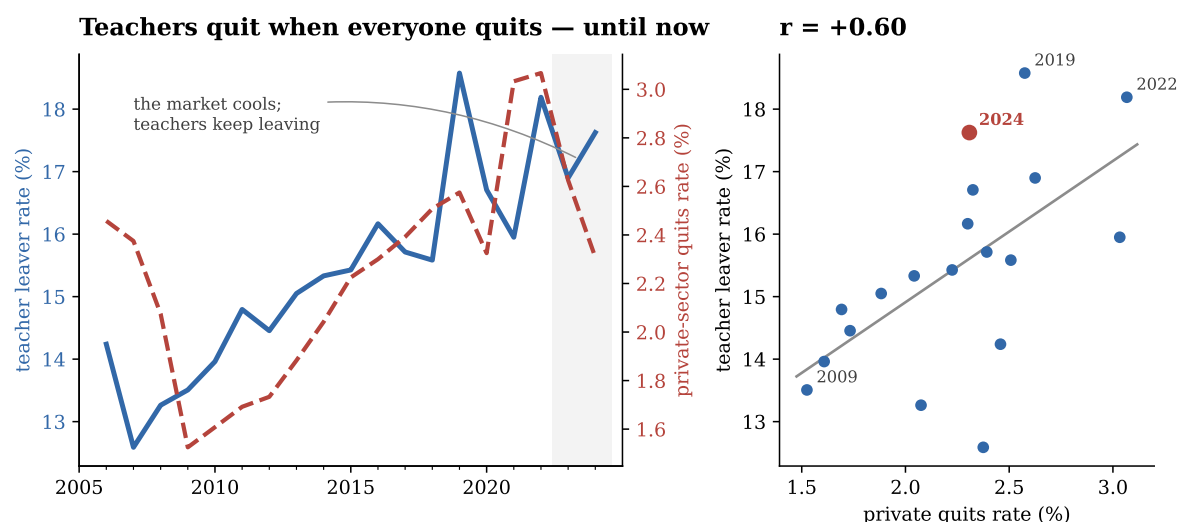


Figure 6: Teacher attrition and the outside labour market.

Left: teacher leaver rate (CPS panel) and the private-sector quits rate (JOLTS, annual average of monthly rates, FRED JTS1000QUR). Right: the same two series as a scatter with an OLS fit; at the 2024 quits rate the fit predicts 15.6 per cent attrition against an observed 17.6. The 2019 point links into 2020 and carries the COVID shock.

6. What this means

EPI closes by warning England that a borrowed recovery “may be susceptible to rapid reversal”: when the graduate market reheats, the underlying pay discount will reassert itself. The American version of the warning is harsher, because the experiment has already run. The market cooled, and retention did not recover — so the United States cannot even wait out the cycle. Its teacher problem is now demonstrably structural: a workforce that shrank 3.7 per cent in one year while pupils also declined, an inflow whose median age is 42, early-career attrition at record highs, and a pay position eight points further behind graduates than in 2010.

The policy translation writes itself, and it is the same on both sides of the Atlantic: what the labour market will not lend, pay and conditions must buy. England’s data show what a lucky macro draw can do for retention; America’s data show what happens without one. Neither is a strategy.

Data and methods. Teacher series: CPS basic monthly microdata 2005–2025 (Census Bureau; NBER mirror before 2021). The 4-8-4 rotation re-interviews each address twelve months apart; persons are linked on household and line identifiers and validated on sex, race and age progression (Madrian–Lefgren). Universe: employed full-time K–12 teachers — occupations 2310 (elementary/middle), 2320 (secondary), 2330 (special education) — with a bachelor’s degree or higher; 148,390 linked teacher-years. All estimates use final person weights. Attrition: share not teaching at $t+12$ (“persistent”: nor at any later interview). Earnings: usual weekly earnings in outgoing rotations, CPI-U-deflated. External series: NCES Digest table 203.10 (enrolment); FRED LEU0252918500, LEU0252881600 (comparator earnings), JTS1000QUR (quits), LNS14027662, CPIAUCSL. No CPS was collected in October 2025 (federal shutdown): 2025 stocks average eleven months and the 2024 base year has no October links. **What the CPS cannot replicate** from England’s School Workforce Census: subject-level out-of-field teaching (see NCES’s NTPS), vacancy counts (see JOLTS state-and-local education), and long-run cohort retention — the rotation caps observation at sixteen months, so Figures 3–6 stand in for EPI’s cohort-survival chart. CPS occupation-based stocks include private-school teachers; NCES enrolment is public-school only, hence both are shown as indices. Code: `src/30_borrowed_recovery_data.py`, `src/31_borrowed_recovery_figures.py` in the project repository.

References

- Education Policy Institute (2026). *A borrowed recovery: the 2025 school workforce data*. EPI, London. <https://epi.org.uk/publications-and-research/a-borrowed-recovery-the-2025-school-workforce-data/>
- Durán Roa, J. E. (2026). *Who Leaves Teaching, and Why? The Profile of Teacher Attrition in the United States, 2005–2025*. Mimeo, University of Alicante (companion paper, this repository).
- Madrian, B. and L. Lefgren (2000). An approach to longitudinally matching Current Population Survey respondents. *Journal of Economic and Social Measurement* 26(1), 31–62.
- Allegretto, S. (2024). *Teacher pay penalty still looms large*. Economic Policy Institute, Washington DC.
- U.S. Bureau of Labor Statistics: Current Population Survey and JOLTS series via FRED. National Center for Education Statistics: Digest of Education Statistics (2025), table 203.10.